Stories We Tell: Narrative Universals and Biases Across Cultures and Language Models

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Abstract

This work asks how storytelling patterns vary across cultures and how language models read and reproduce them. Sixty stories from three traditions (European literary fiction, Japanese folktales, South African folktales) are modelled as typed event graphs: five key events per story, each labelled with one of eleven narrative functions inspired by Propp and Campbell. Two free LLMs, Llama 3.1 8B and Gemini 2.5 Flash, annotate the stories, and their output is checked against LitBank, a spaCy baseline, a replica run and an independent manual annotation. Both models find a mostly shared structure (mean similarity **0.205**) with a small but significant tradition effect, one culture-bound function (Trickster, present in **80%** to **90%** of South African tales) and two universal ones (Call to adventure, Ordeal). The arc they assign depends on the model: European fiction is Tragedy for one and Hero's journey for the other. In a blind retelling experiment, no story retold in another tradition becomes closer to the target culture, and even retellings within the original tradition fall below the annotator noise floor. The models keep the structural skeleton and change the surface.

Keywords: narrative structure, large language models, cultural bias, narrative universals, Propp functions, graph similarity

1 Introduction

Stories carry values from one generation to the next. Propp reduced Russian wondertales to a fixed sequence of 31 functions [1], and Campbell proposed one cross-cultural pattern, the hero's journey [2]. It is still debated whether such patterns are universal or products of the corpora and analysts that produced them. Language models now

summarise, retell and generate stories, so a second question follows: what do they read into a story from another culture, and what do they keep when asked to retell it in a different one?

Computational work on narrative already covers parts of this. Valls-Vargas et al. predict Proppian functions from stories in natural language [3], Kumaran et al. use LLMs to generate partially ordered event graphs for interactive stories [4], and Ranade et al. survey computational narrative understanding [5]. Gold entity annotations of literature exist [6] and graph comparison has well-established tools [7, 8]. The protocol presented here is not based on any single study that puts these pieces together into a validated, cross-cultural procedure that also tests LLM retellings. Cultural bias is treated as an empirical question, with the results of the study itself as the evidence.

Declaration on the use of generative AI.. Generative AI was used in two ways. First, as objects of the research: Llama 3.1 8B (run locally through Ollama) and Gemini 2.5 Flash (Google AI Studio, free tier) annotated the corpus and generated the retellings, as described in Sect. 2 and Sect. 3.1. Second, as assistants in the work: Claude by Anthropic, through the Claude Code environment (models Opus 5.5 and Sonnet 5.5), was used to write and restructure the Python code of the pipeline, its tests and the notebook of checks; to draft and rewrite the text of this paper, the project documentation and the LaTeX source. The author supplied the task, the corpus and its sources, decided which results to report, and annotated the manual sheet of 12 stories by hand. All numbers in the paper are computed by the project code from the saved annotations and can be checked with the accompanying notebook. Text and code produced with AI assistance were edited by the author, executed and tested (offline test suite and notebook of checks), and the author is responsible for the final content.

2 Research question and methodology

Problem.

Let $\mathcal{S}$ be a set of stories, each belonging to a tradition $c(s) \in \{\text{European}, \text{Asian}, \text{South African}\}$. An annotator A (an LLM) maps a text s to an event graph $G_s = A(s)$.

Definition 1 (Event graph) A typed directed multigraph with CHARACTER nodes (role: protagonist, antagonist, deuteragonist, helper, trickster, secondary), exactly five ordered EVENT nodes (each labelled by one of eleven narrative functions) and LOCATION nodes. Edges are *precedes*, *causes*, *participates*, *takes_place_at*, *transforms_in* and typed character relations.

A similarity $\sigma(G_s, G_t) \in [0, 1]$ induces the distance $D = 1 - \sigma$. The study addresses three questions: (RQ1) are stories closer within than between traditions, and which functions are universal or culture-bound; (RQ2) how much do these conclusions, and the arc assigned to each tradition, depend on the annotating model; (RQ3) when a generator retells s in a target tradition, is the structure preserved, assimilated to the target, or distorted beyond what annotator noise and paraphrase explain.

Proposed solution.

An LLM extracts entities and events, because classical NLP tools (NER, parsing) find names and places but cannot assign narrative functions or causal links. Each text is annotated in one call with a controlled vocabulary of five arcs and eleven functions, each defined operationally in the prompt (for example ORDEAL: "a struggle, trial, difficult task or confrontation with the antagonist"). Long texts are reduced to three windows (beginning, middle, end), the title is hidden and outside knowledge is forbidden. Outputs are checked against a schema. A spaCy NER baseline [9], used only for validation, and LitBank gold entities [6] check the extraction.

Metrics.

Four structural similarities, each catching what the others miss: approximate graph edit distance [7] (order and topology), motif overlap (Jaccard index over unigrams, temporal bigrams and causal pairs of functions), semantic role alignment (Jaccard over role and function pairs and over role, relation, role triples) and Weisfeiler–Lehman histograms [8] ($h = 3$, labels with neighbourhood structure). The composite is their mean,

$$\sigma = \tfrac{1}{4}\left(\sigma_{\mathrm{GED}} + \sigma_{\mathrm{motif}} + \sigma_{\mathrm{role}} + \sigma_{\mathrm{WL}}\right), \tag{1}$$

and a TF-IDF/LSA cosine on event descriptions gives a separate semantic similarity, so that sharing a structure and sharing a content stay apart. Table 1 lists every metric, what it is used for and how to read its values.

Statistics.

The tradition effect is $\Delta = \overline{D}_{\mathrm{between}} - \overline{D}_{\mathrm{within}}$, tested by permuting tradition labels 5000 times. Each function's share per story is compared across traditions with Kruskal–Wallis [10] and Benjamini–Hochberg correction [11]: a function is *culture-bound* if significant, *universal* if frequent everywhere without significant differences. Annotators are compared with Cohen's κ [12].

Blind retelling.

A generator receives only the text and the name of a target tradition, never the arc or the functions to preserve, and the corpus annotator annotates the result. Two controls calibrate the drift: a paraphrase control (a retelling in the story's own tradition) and a noise floor (a re-annotation of the original at temperature 0.5). A retelling is *preserved* if it is at least as similar to the original as the paraphrase control, up to a tolerance; otherwise it is *assimilated* (closer to the target tradition's corpus), *hybrid* (equidistant) or *distorted*.

3 Experimental results

3.1 Dataset and setup

The corpus has 60 English texts, 20 per tradition: LitBank novels [6] (European), Japanese folktales (Asian, collected from the web resource Il Bazar di Mari [13]) and South African folktales (from the Sacred Texts collection [14]). Some texts may also

Table 1 Metrics used, their purpose and how to read the values. All similarities lie in $[0, 1]$.

Metric	Used for	Reading
GED similarity	Edit cost to turn one event graph into the other, divided by graph size	0 = nothing in common, 1 = identical graphs
Motif overlap	Shared function sequences and causal pairs	0 = no shared plot pattern, 1 = same patterns
Role alignment	Shared (role, function) pairs and role relations	0 = different casts and roles, 1 = same
Weisfeiler–Lehman	Labels together with neighbourhood structure	0 = no common local structure, 1 = same
Composite σ	Mean of the four above; one number per story pair	In this corpus: 0.205, i.e. low overlap
Semantic similarity	TF-IDF/LSA cosine of event descriptions	0 = unrelated content, 1 = same content
Tradition effect Δ	Mean between-tradition distance minus within	> 0: closer within a tradition; ≈ 0: no effect
Kruskal–Wallis q	Does a function's share differ across traditions?	$q < 0.05$: culture-bound function
Cohen's κ	Agreement between two annotations beyond chance	0 = chance, 1 = perfect; 0.4 to 0.6 is moderate
Recall@5	Top-5 gold character names found by the annotator	0 = none found, 1 = all found
Grounding	Extracted characters that appear in the text	1 = every character is in the text
Retention / intrusion	Functions of the original kept in the retelling / added by it	1 = all kept (retention); high intrusion = invented beats
Noise and paraphrase floor	Similarity of the original to its own re-annotation, and to a same-tradition retelling	Reference levels: a retelling is preserved only if it reaches the paraphrase floor

have been translated into English as a second step, after they were obtained from their source, to get one uniform language across the corpus. Together with the translations already present in the sources, this confounds translator and tradition effects. The annotators are Llama 3.1 8B [15] (local, through Ollama) and Gemini 2.5 Flash [16] (free tier), both at temperature 0. The retelling sample is seeded: three stories per tradition, each retold into four targets (European, East Asian, Southern African, Indigenous American) and annotated by Llama.

3.2 Validity of the annotators

Table 2 summarises the checks. Extracted characters appear in the text in 97.6% to 100% of cases. Recall@5 on LitBank is below the spaCy baseline, but recall does not penalise precision and the gold covers only the first ∼2000 words of each novel, so it is read as a plausibility check. The arc label is stable within a model (replica $\kappa = 0.97$) and not across models ($\kappa = 0.26$). Event functions are noisy even within a model ($\kappa = 0.45$), about as noisy as the disagreement between models ($\kappa = 0.36$), yet the annotation of the same story is the closest one in 20% of cross-model and 58% of replica comparisons (chance 1.7%). A manual annotation of 12 stories (4 per tradition), made by the author independently of the two LLM annotators, agrees moderately with both models on functions, and with Gemini more than with Llama on the arc (8 against 5 of 12); with so few stories this is only indicative.

Table 2 Validation of the annotators ($n = 60$ stories; manual sheet $n = 12$).

Measure	Llama 3.1 8B	Gemini 2.5 Flash
Grounding (Eur./Asia/S. Afr.)	.98/1.00/.99	.98/.99/1.00
LitBank recall@5 (spaCy .48)	.39	.44
Manual sheet: arc agreement	.42	.67
Manual sheet: function agreement / κ	.55 / .50	.45 / .37
Between models: arc agr. / κ	.42 / .26	
Between models: function κ	.36	
Replica (Llama): arc agr. / κ	.98 / .97	
Replica (Llama): function κ	.45	

3.3 RQ1 and RQ2: shared structure, culture-bound functions, model dependence

Mean composite similarity is 0.205 for both models (GED .175/.169, motif .168/.161, role .171/.165, WL .307/.326 for Llama/Gemini), so most stories differ in exact event structure whatever their tradition. Stories are still closer within than between traditions ($\Delta = 0.016$ for Llama, 0.034 for Gemini, both $p < 0.001$). South African tales drive the effect ($\Delta = 0.020/0.036$ against Asian and $0.030/0.058$ against European stories, all $p < 0.001$); Asian and European stories cannot be told apart for Llama ($\Delta = -0.003$, $p = 0.76$) and are barely separated for Gemini (0.007, $p = 0.043$). Clustering the same distances is unstable (degenerate or weakly aligned with tradition, ARI ≤ 0.15), so no conclusion is drawn from it.

Table 3 gives the share of stories containing each function. Trickster is the only culture-bound function for both annotators ($q < 0.001$): it occurs in 80% to 90% of South African tales and in 15% to 35% of the others. Call to adventure and Ordeal appear in most stories of every tradition, which fits a shared Propp and Campbell scaffold. Helper appears is universal for Llama and not for Gemini.

Table 3 Share of stories with each function by tradition (Asia/Europe/S. Africa); BH-adjusted q in brackets for significant tests.

Function	Llama 3.1 8B	Gemini 2.5 Flash	Label
Trickster	.35/.15/.80 ($q < .001$)	.35/.25/.90 ($q < .001$)	culture-bound
Call to adventure	.85/.85/.90	.90/1.00/.90	universal
Ordeal	.60/.50/.60	.70/.85/.65	universal
Helper appears	.90/.85/.55	.55/.35/.40	model-dependent

The model also changes how a whole tradition is read (Fig. 1). Llama labels 13/20 European texts as Tragedy and 19/20 South African tales as Trickster tale; Gemini labels 16/20 European texts as Hero's journey and 14/20 South African tales as Trickster tale, and gives Cautionary tale to 8/20 Asian texts where

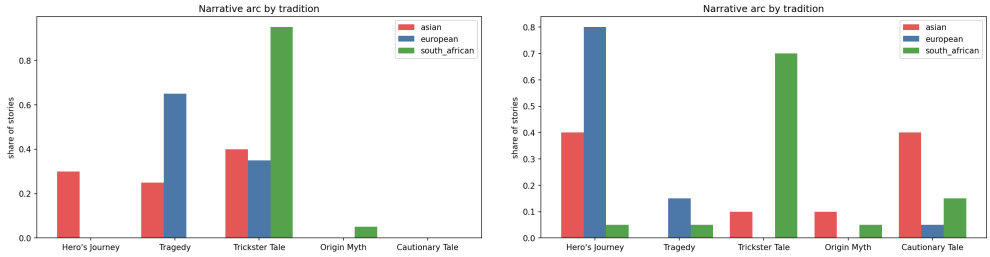


Fig. 1 Arc labels per tradition. Left: Llama 3.1 8B; right: Gemini 2.5 Flash.

Llama gives it to none. The conclusions on TRICKSTER, CALL TO ADVENTURE and ORDEAL hold across models; those on the arc of a tradition describe the annotator as much as the corpus.

3.4 RQ3: retelling across cultures

Llama produced 36 retellings (27 cross-tradition and 9 paraphrase controls) and Gemini 20 (15 cross and 5 controls). Each retelling is judged against the paraphrase floor of its story (Table 4).

Table 4 Outcomes of cross-tradition retellings. "Sim." is the mean composite similarity to the original. No retelling was classified as assimilated. The annotator noise floor is 0.42.

Generator (stories)	Cross	Preserved	Hybrid	Distorted	Arc kept	Sim. cross / control
Llama 3.1 8B (9)	27	18	7	2	.41	.306 / .306
Llama 3.1 8B (same 5)	15	10	4	1	.13	.314 / .288
Gemini 2.5 Flash (5)	15	5	6	4	.40	.296 / .370

For Llama, a retelling into another tradition is exactly as similar to the original as a paraphrase within the original tradition (0.306 against 0.306; paired Wilcoxon over nine stories, $p = 1.0$), and none ends up closer to the target tradition's corpus than to the source's. Semantic similarity is only 0.19 for Llama and 0.07 for Gemini, against structural similarity around 0.3: the surface changes while the structure largely stays. Retention is highest for CALL TO ADVENTURE (1.00 for Llama, 0.87 for Gemini) and HELPER APPEARS (0.81 and 0.73), while SACRIFICE and RETURN are lost; ORDEAL and MORAL REWARD show the highest intrusion, so retellings add conventional plot beats. The generators differ: on the five stories both completed, Llama preserves 10/15 retellings and Gemini 5/15, and Gemini's cross-tradition retellings are less similar to the original than its own controls (0.296 against 0.370; with fewer than six pairs no test was run).

4 Concluding remarks

At the level of five typed events, narrative structure is largely shared by the three traditions. The tradition effect is small and sits mostly in the South African tales, whose distinctive trait is the trickster; Call to adventure and Ordeal behave as universals and Trickster as the only culture-bound function. These comparisons depend on the LLM that annotates: both models agree on the function-level findings and little on individual labels, and the arc of a tradition changes with the model. In retelling, the models keep the shared skeleton but do not adapt the plot to the target culture. Even the retellings within the original tradition fall below the annotator noise floor (0.31 against 0.42), so no regenerated story reproduces exactly the cultural sphere of the original, and none moves to the sphere of the target.

These conclusions have limits. The LLM is both instrument and object of study, and function labels are noisy across replicas ($\kappa = 0.45$). Five events per story is a coarse abstraction, and novels are represented by three windows. Translation, including possible secondary translation after acquisition, confounds translator and tradition, and some South African tales share a tale type. The retelling sample is small and annotator noise is of the same order as the drift measured. Future work should add independent multi-rater human annotation, larger models and original-language texts, a fourth tradition, and retellings that state the target tradition's narrative schema explicitly.

Data and code availability. https://github.com/denisguareschi/StoriesWeTellProject

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